CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Linear Transformations

Worked solutions 11 questions · 85 marks

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1. The transformation U, represented by the 2×2 matrix P, is a reflection in the x-axis.

(a) Write down the matrix P. **[1]**

The transformation V, represented by the 2×2 matrix Q, is a reflection in the line $y = x$.

(b) Write down the matrix Q. **[1]**

Given that U followed by V is transformation T, which is represented by the matrix R

(c) express R in terms of P and Q, **[1]**

(d) find the matrix R, **[2]**

(e) give a full geometrical description of T as a single transformation. **[2]**

Solution

(a) A reflection in the x-axis maps (x, y) to $(x, -y)$, so

$$P = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

(b) A reflection in the line $y = x$ swaps the coordinates, so

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(c) If a point is first transformed by U and then by V, then with column vectors

$$\mathbf{x} \mapsto P\mathbf{x} \mapsto Q(P\mathbf{x}) = (QP)\mathbf{x}$$

Hence

$$R = QP$$

(d) Using part (c),

$$R = QP = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Multiplying,

$$R = \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot (-1) \\ 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

(e) Apply R to a general point:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

So $(x, y) \mapsto (-y, x)$.

This is a rotation through $90°$ anticlockwise about the origin.

Therefore T is a rotation of $90°$ anticlockwise about the origin.

2. A right-angled triangle T has vertices $A(-2,1)$, $B(1,1)$ and $C(1,5)$. When T is transformed by the matrix

$$P = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

the image is T'.

(a) Find the coordinates of the vertices of T'. **[2]**

(b) Describe fully the transformation represented by P. **[2]**

The matrices

$$Q = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$$

and

$$R = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$$

represent two transformations. When T is transformed by the matrix QR, the image is T''.

(c) Find QR. **[2]**

(d) Find the determinant of QR. **[2]**

(e) Using your answer to part (d), find the area of T''. **[3]**

Solution

(a) A point (x, y) is represented by the column vector $\begin{pmatrix} x \\ y \end{pmatrix}$.

Applying

$$P = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

gives

$$P\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

So each point (x, y) maps to $(-y, x)$.

For $A(-2,1)$,

$$A' = (-1,-2)$$

For $B(1,1)$,

$$B' = (-1,1)$$

For $C(1,5)$,

$$C' = (-5,1)$$

Hence the vertices of T' are $A'(-1,-2)$, $B'(-1,1)$, $C'(-5,1)$.

(b) To identify the transformation, look at what P does to the unit basis vectors.

$$P\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}$$

so the positive x-axis maps to the positive y-axis.

Also,

$$P\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\end{pmatrix}$$

so the positive y-axis maps to the negative x-axis.

This is a rotation of $90°$ anticlockwise about the origin.

(c)

$$QR = \begin{pmatrix}2 & -1\\0 & 3\end{pmatrix}\begin{pmatrix}1 & 2\\1 & 0\end{pmatrix}$$

Multiplying row by column:

$$QR = \begin{pmatrix}2(1)+(-1)(1) & 2(2)+(-1)(0)\\0(1)+3(1) & 0(2)+3(0)\end{pmatrix} = \begin{pmatrix}1 & 4\\3 & 0\end{pmatrix}$$

(d) Using

$$QR = \begin{pmatrix}1 & 4\\3 & 0\end{pmatrix}$$

its determinant is

$$\det(QR) = 1\cdot 0 - 4\cdot 3 = -12$$

(e) The original triangle T is right-angled with side lengths

$$AB = 3, \qquad BC = 4$$

so its area is

$$\text{Area of } T = \frac{1}{2}\times 3\times 4 = 6$$

The magnitude of the determinant gives the area scale factor, so

$$\text{area scale factor} = |\det(QR)| = |-12| = 12$$

Therefore,

$$\text{Area of } T'' = 12\times 6 = 72$$

3. **(i)** In each of the following cases, find a 2×2 matrix that represents

(a) a reflection in the line $y = x$ **[1]**

(b) a rotation of $60°$ clockwise about $(0, 0)$ **[1]**

(c) a rotation of $60°$ clockwise about $(0, 0)$ followed by a reflection in the line $y = x$ **[2]**

(ii) The triangle T has vertices at the points $(1, 2)$, $(5, 5)$ and $(k, -1)$, where $k > -3$ is a constant.

Triangle T is transformed onto the triangle T' by the matrix

$$\begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix}$$

Given that the area of triangle T' is 84 square units, find the value of k. **[6]**

Solution

(i) For a 2×2 transformation matrix, the first column is the image of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and the second column is the image of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

(a) A reflection in the line $y = x$ swaps the coordinates, so

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Hence the matrix is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(b) A clockwise rotation through angle θ about the origin has matrix

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

With $\theta = 60°$,

$$\cos 60° = \frac{1}{2}, \qquad \sin 60° = \frac{\sqrt{3}}{2}$$

So the matrix is

$$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

(c) Let

$$R = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Since the rotation is done first and the reflection second, the required matrix is FR.

$$FR = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 0 \cdot \frac{1}{2} + 1 \cdot \left(-\frac{\sqrt{3}}{2}\right) & 0 \cdot \frac{\sqrt{3}}{2} + 1 \cdot \frac{1}{2} \\ 1 \cdot \frac{1}{2} + 0 \cdot \left(-\frac{\sqrt{3}}{2}\right) & 1 \cdot \frac{\sqrt{3}}{2} + 0 \cdot \frac{1}{2} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

(ii) Let

$$A = \begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix}$$

The determinant of A is

$$\det(A) = 2(-1) - 5(1) = -7$$

A transformation by a matrix scales area by the factor $|\det(A)|$, so the area scale factor is

$$|-7| = 7$$

Hence the area of triangle T is

$$\frac{84}{7} = 12$$

Now find the area of T in terms of k.

Using the vertex $(1, 2)$, the side vectors are

$$\begin{pmatrix} 5-1 \\ 5-2 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \qquad \begin{pmatrix} k-1 \\ -1-2 \end{pmatrix} = \begin{pmatrix} k-1 \\ -3 \end{pmatrix}$$

So

$$\text{Area of } T = \frac{1}{2}\left|\det\begin{pmatrix} 4 & k-1 \\ 3 & -3 \end{pmatrix}\right|$$

$$= \frac{1}{2}|4(-3) - 3(k-1)|$$

$$= \frac{1}{2}|-12 - 3k + 3|$$

$$= \frac{1}{2}|-3k - 9|$$

$$= \frac{3}{2}|k + 3|$$

Since $k > -3$, we have $k + 3 > 0$, so

$$|k + 3| = k + 3$$

Therefore

$$\frac{3}{2}(k + 3) = 12$$

$$k + 3 = 8$$

$$k = 5$$

Hence the value of k is 5.

4.

$$\mathbf{P} = \begin{pmatrix} p+1 & 2 \\ 2 & p+1 \end{pmatrix}$$

where p is a positive constant.

The matrix $\mathbf{P}$ represents a linear transformation U.
The parallelogram S has vertices at the points with coordinates $(1,1)$, $(4,2)$, $(6,5)$ and $(3,4)$.
The area of the image of S under the linear transformation U is 35.

(a) Determine the value of p. **[4]**

The transformation V consists of a reflection in the line $y = x$ followed by a stretch of scale factor 2 parallel to the x-axis with the y-axis invariant. The transformation V is represented by the matrix $\mathbf{Q}$.

(b) Write down the matrix $\mathbf{Q}$. **[2]**

Given that U followed by V is the transformation W, which is represented by the matrix $\mathbf{R}$

(c) find the matrix $\mathbf{R}$. **[2]**

Solution

(a) Take two adjacent sides of the parallelogram from the vertex $(1,1)$:

$$\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \qquad \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

So the area of S is

$$\left| \det \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix} \right| = |3 \cdot 3 - 2 \cdot 1| = |9 - 2| = 7$$

A linear transformation represented by matrix $\mathbf{P}$ multiplies area by the factor $|\det \mathbf{P}|$.

Since the image of S has area 35,

$$35 = |\det \mathbf{P}| \times 7$$

so

$$|\det \mathbf{P}| = 5$$

Now

$$\mathbf{P} = \begin{pmatrix} p+1 & 2 \\ 2 & p+1 \end{pmatrix}$$

so

$$\begin{aligned} \det \mathbf{P} &= (p+1)(p+1) - 2 \cdot 2 \\ &= (p+1)^2 - 4 \\ &= p^2 + 2p + 1 - 4 \\ &= p^2 + 2p - 3 \end{aligned}$$

Hence

$$|p^2 + 2p - 3| = 5$$

So either

$$p^2 + 2p - 3 = 5$$

or

$$p^2 + 2p - 3 = -5$$

From the first:

$$p^2 + 2p - 8 = 0$$

$$(p + 4)(p - 2) = 0$$

so

$$p = -4 \quad \text{or} \quad p = 2$$

From the second:

$$p^2 + 2p + 2 = 0$$

which has no real solutions.

Since p is positive,

$$p = 2$$

(b) Reflection in the line $y = x$ has matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

A stretch of scale factor 2 parallel to the x-axis, with the y-axis invariant, has matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

Since the reflection is followed by the stretch,

$$\mathbf{Q} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

(c) From part (a), $p = 2$, so

$$\mathbf{P} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$$

Since U followed by V is represented by $\mathbf{R}$,

$$\mathbf{R} = \mathbf{QP}$$

Therefore

$$\begin{aligned}\mathbf{R} &= \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 0\cdot 3 + 2\cdot 2 & 0\cdot 2 + 2\cdot 3 \\ 1\cdot 3 + 0\cdot 2 & 1\cdot 2 + 0\cdot 3 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 6 \\ 3 & 2 \end{pmatrix}\end{aligned}$$

5. Consider the matrix

$$A = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

(i) Calculate A^2. **[1]**

You are told that the matrix A represents a reflection in a line through the origin.

(ii) Explain why your answer to part (i) is consistent with this information. **[1]**

(iii) By finding the invariant points of the transformation, determine the equation of the mirror line. **[3]**

(iv) Describe fully the transformation represented by

$$B = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$ **[2]**

(v) A composite transformation is formed by the transformation represented by A followed by the transformation represented by B. Find the single matrix that represents this composite transformation. **[2]**

(vi) The composite transformation described in part (v) is equivalent to a single reflection. Find the equation of the mirror line of this reflection. **[1]**

Solution

(i)

$$A^2 = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

Calculating each entry,

$$A^2 = \begin{pmatrix} \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right) \\ \left(\frac{\sqrt{3}}{2}\right)\left(-\frac{1}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{4} + \frac{3}{4} & -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & \frac{3}{4} + \frac{1}{4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

(ii) A reflection done twice takes every point back to its original position, so its matrix squared should be the identity matrix.

Since in part (i) we found $A^2 = I$, this is consistent with A representing a reflection.

(iii) Points on the mirror line are unchanged by the reflection, so invariant points satisfy

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Thus

$$\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

This gives the simultaneous equations

$$-\frac{1}{2}x + \frac{\sqrt{3}}{2}y = x$$

$$\frac{\sqrt{3}}{2}x + \frac{1}{2}y = y$$

Multiply both equations by 2:

$$-x + \sqrt{3}\,y = 2x$$

$$\sqrt{3}\,x + y = 2y$$

So

$$-3x + \sqrt{3}\,y = 0$$

$$\sqrt{3}\,x - y = 0$$

From either equation,

$$y = \sqrt{3}\,x$$

So the mirror line is

$$y = \sqrt{3}x$$

(iv) The standard matrix for a rotation through angle θ about the origin is

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Now

$$B = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

So

$$\cos\theta = -\frac{1}{2}, \qquad \sin\theta = \frac{\sqrt{3}}{2}$$

This gives

$$\theta = 120^\circ$$

Therefore B represents a rotation of 120° anticlockwise about the origin.

(v) If A is followed by B, the composite matrix is BA.

$$BA = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

$$= \begin{pmatrix} \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right) + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right) \\ \left(\frac{\sqrt{3}}{2}\right)\left(-\frac{1}{2}\right) + \left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2}\right)\left(\frac{1}{2}\right) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{4} - \frac{3}{4} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} & \frac{3}{4} - \frac{1}{4} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

(vi) Let

$$M = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

The mirror line consists of invariant points, so solve

$$M\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

This gives

$$-\frac{1}{2}x - \frac{\sqrt{3}}{2}y = x$$

$$-\frac{\sqrt{3}}{2}x + \frac{1}{2}y = y$$

Multiplying by 2,

$$-x - \sqrt{3}\,y = 2x$$

$$-\sqrt{3}\,x + y = 2y$$

So

$$-3x - \sqrt{3}\,y = 0$$

$$-\sqrt{3}\,x - y = 0$$

Hence

$$y = -\sqrt{3}x$$

So the mirror line is

$$y = -\sqrt{3}x$$

6. The matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$

where $k > 0$ and $k \neq 1$.

(a) The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations. State the type of each transformation, and make clear the order in which they are applied. **[2]**

The unit square in the x-y plane is transformed by $\mathbf{M}$ onto rectangle $OABC$.

(b) Find, in terms of k, the area of rectangle $OABC$ and the matrix which transforms $OABC$ onto the unit square. **[3]**

(c) Show that the line through the origin with gradient $\frac{1}{\sqrt{k}}$ is invariant under the transformation in the x-y plane represented by $\mathbf{M}$. **[3]**

Solution

(a) Write

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$

The right-hand matrix is applied first. Since

$$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ ky \end{pmatrix}$$

this is a stretch parallel to the y-axis, with scale factor k.

Then

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$$

which swaps the coordinates, so this is a reflection in the line $y = x$.

Therefore the transformations are: first a stretch in the y-direction by scale factor k, then a reflection in the line $y = x$.

(b) First find $\mathbf{M}$:

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$$

The area scale factor of a transformation is $|\det \mathbf{M}|$, so

$$\det \mathbf{M} = \begin{vmatrix} 0 & k \\ 1 & 0 \end{vmatrix} = 0 - k = -k$$

Hence the area scale factor is

$$|\det \mathbf{M}| = k$$

Since the unit square has area 1, the area of rectangle $OABC$ is

$$1 \times k = k$$

To transform $OABC$ back to the unit square, we need $\mathbf{M}^{-1}$. For

$$\mathbf{M} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$$

$$\mathbf{M}^{-1} = \frac{1}{-k}\begin{pmatrix} 0 & -k \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{1}{k} & 0 \end{pmatrix}$$

Therefore the area of $OABC$ is k, and the required matrix is

$$\begin{pmatrix} 0 & 1 \\ \frac{1}{k} & 0 \end{pmatrix}$$

(c) The line through the origin with gradient $\frac{1}{\sqrt{k}}$ has equation

$$y = \frac{x}{\sqrt{k}}$$

Take any point on this line:

$$\begin{pmatrix} x \\ \dfrac{x}{\sqrt{k}} \end{pmatrix}$$

Apply $\mathbf{M} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$:

$$\mathbf{M}\begin{pmatrix} x \\ \dfrac{x}{\sqrt{k}} \end{pmatrix} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ \dfrac{x}{\sqrt{k}} \end{pmatrix} = \begin{pmatrix} \dfrac{kx}{\sqrt{k}} \\ x \end{pmatrix} = \begin{pmatrix} \sqrt{k}\,x \\ x \end{pmatrix}$$

Let the image point be $(X, Y) = (\sqrt{k}\,x,\ x)$. Then

$$\frac{X}{\sqrt{k}} = \frac{\sqrt{k}\,x}{\sqrt{k}} = x = Y$$

so

$$Y = \frac{X}{\sqrt{k}}$$

Therefore the image point also lies on the line $y = \frac{x}{\sqrt{k}}$.

Hence the line through the origin with gradient $\frac{1}{\sqrt{k}}$ is invariant under the transformation represented by $\mathbf{M}$.

7. Bilal has been asked to describe the transformation given by the matrix

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$$

He writes his answer as follows:

> The transformation is a rotation about the y-axis through an angle of θ, where
>
> $$\cos\theta = \frac{\sqrt{3}}{2} \quad \text{and} \quad \sin\theta = \frac{1}{2}$$
>
> $$\theta = 30^\circ$$

Identify and correct the error in Bilal's work. **[2]**

Solution

The standard matrix for a rotation about the y-axis is

$$R_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

Comparing this with

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$$

gives

$$\cos\theta = \frac{\sqrt{3}}{2}, \qquad \sin\theta = -\frac{1}{2}$$

So the angle is not 30°, but

$$\theta = -30^\circ$$

equivalently 330°.

Bilal's error is that he ignored the negative sign on the sine term. The transformation is a rotation about the y-axis through 30° clockwise when viewed from the positive y-axis.

8. You are given the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

(a) Find $\mathbf{A}^5$. **[1]**

(b) Describe the transformation that $\mathbf{A}$ represents. **[2]**

The matrix $\mathbf{B}$ represents a reflection in the plane $x = y$.

(c) Write down the matrix $\mathbf{B}$. **[1]**

The point Q has coordinates $(4, -1, 2)$. The point Q' is the image of Q under the transformation represented by $\mathbf{B}$.

(d) Find the coordinates of Q'. **[1]**

Solution

(a) First find $\mathbf{A}^2$:

$$\mathbf{A}^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Now multiply once more:

$$\mathbf{A}^3 = \mathbf{A}^2\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}$$

So

$$\mathbf{A}^5 = \mathbf{A}^3\mathbf{A}^2 = \mathbf{I}\mathbf{A}^2 = \mathbf{A}^2$$

Hence

$$\mathbf{A}^5 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

(b) Let a general point be $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y \\ z \\ x \end{pmatrix}$$

So $\mathbf{A}$ sends (x, y, z) to (y, z, x), which is a cyclic permutation of the coordinates.

Also, any point on the line $x = y = z$ is unchanged, so this is a rotation about the line $x = y = z$.

Since $\mathbf{A}^3 = \mathbf{I}$, three applications give a full turn, so the angle of rotation is

$$\frac{360^\circ}{3} = 120^\circ$$

For the sense, $\mathbf{A}$ sends $\mathbf{i} \mapsto \mathbf{k}$, $\mathbf{k} \mapsto \mathbf{j}$ and $\mathbf{j} \mapsto \mathbf{i}$. Viewed from $(1,1,1)$ towards the origin, the positive x-, y- and z-axes appear in anticlockwise order, so $x \to z \to y$ is clockwise.

Therefore $\mathbf{A}$ represents a rotation of $120°$ clockwise about the line $x = y = z$, viewed from $(1,1,1)$ towards the origin.

(c) A reflection in the plane $x = y$ swaps the x- and y-coordinates and leaves z unchanged.

Therefore

$$\mathbf{B} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(d) Apply $\mathbf{B}$ to $Q = (4, -1, 2)$:

$$\mathbf{Q}' = \mathbf{B}\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$$

So the coordinates of Q' are $(-1, 4, 2)$.

9. Given that the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

represents a rotation through $\alpha°$ about the y-axis in the positive sense given by the right-hand rule

(a) determine the value of α. **[1]**

(b) Hence determine the smallest possible positive integer value of k for which $\mathbf{M}^k = \mathbf{I}$. **[2]**

The 3×3 matrix $\mathbf{N}$ represents a reflection in the plane with equation $x = 0$.

(c) Write down the matrix $\mathbf{N}$. **[1]**

The point A has coordinates $(1, 2, 3)$.

The point B is the image of the point A under the transformation represented by matrix $\mathbf{M}$ followed by the transformation represented by matrix $\mathbf{N}$.

(d) Show that the coordinates of B are $(2\sqrt{2}, 2, -\sqrt{2})$. **[2]**

Given that O is the origin

(e) show that, to 3 significant figures, the size of angle AOB is $79.4°$. **[2]**

(f) Hence determine the area of triangle AOB, giving your answer to 3 significant figures. **[2]**

Solution

(a) Comparing $\mathbf{M}$ with the standard rotation matrix about the y-axis,

$$\begin{pmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{pmatrix}$$

gives

$$\cos\alpha = -\frac{\sqrt{2}}{2}, \qquad \sin\alpha = -\frac{\sqrt{2}}{2}$$

So α is in the third quadrant, hence

$$\alpha = 225°$$

(b) $\mathbf{M}$ represents a rotation of $225°$, so $\mathbf{M}^k$ represents a rotation of $225k°$.

For $\mathbf{M}^k = \mathbf{I}$, the total rotation must be a multiple of $360°$, so

$$225k = 360n$$

for some positive integer n.

Dividing by 45,

$$5k = 8n$$

The smallest positive integer k satisfying this is $k = 8$ (with $n = 5$).

So the smallest possible value of k is 8.

(c) Reflection in the plane $x = 0$ changes (x, y, z) to $(-x, y, z)$, so

$$\mathbf{N} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(d) The point A has position vector

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

First apply $\mathbf{M}$:

$$\mathbf{Ma} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{\sqrt{2}}{2}(1) - \frac{\sqrt{2}}{2}(3) \\ 2 \\ \frac{\sqrt{2}}{2}(1) - \frac{\sqrt{2}}{2}(3) \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{4\sqrt{2}}{2} \\ 2 \\ -\frac{2\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} -2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix}$$

Then apply $\mathbf{N}$:

$$\mathbf{N}(\mathbf{Ma}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} 2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix}$$

Hence the coordinates of B are

$$(2\sqrt{2},\ 2,\ -\sqrt{2})$$

(e) The position vectors of A and B are

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \qquad \overrightarrow{OB} = \begin{pmatrix} 2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix}$$

Using the scalar product,

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = 1(2\sqrt{2}) + 2(2) + 3(-\sqrt{2}) = 4 - \sqrt{2}$$

Also,

$$|\overrightarrow{OA}| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

and

$$|\overrightarrow{OB}| = \sqrt{(2\sqrt{2})^2 + 2^2 + (-\sqrt{2})^2} = \sqrt{8+4+2} = \sqrt{14}$$

So if $\theta = \angle AOB$, then

$$\cos\theta = \frac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|\overrightarrow{OA}|\,|\overrightarrow{OB}|} = \frac{4-\sqrt{2}}{14}$$

Therefore

$$\theta = \cos^{-1}\left(\frac{4-\sqrt{2}}{14}\right) \approx 79.4^\circ$$

So the size of angle AOB is 79.4° to 3 significant figures.

(f) Using the area formula for a triangle,

$$\text{Area of } \triangle AOB = \frac{1}{2}|\overrightarrow{OA}|\,|\overrightarrow{OB}|\,\sin\angle AOB$$

From part (e), $|\overrightarrow{OA}| = |\overrightarrow{OB}| = \sqrt{14}$, so

$$\text{Area} = \frac{1}{2}(\sqrt{14})(\sqrt{14})\sin\theta = 7\sin\theta$$

Using

$$\cos\theta = \frac{4-\sqrt{2}}{14}$$

we get

$$\sin\theta = \sqrt{1-\left(\frac{4-\sqrt{2}}{14}\right)^2}$$

Hence

$$\text{Area} = 7\sqrt{1-\left(\frac{4-\sqrt{2}}{14}\right)^2}$$

$$\approx 6.88$$

So the area of triangle AOB is 6.88 square units to 3 significant figures.

10. Three non-singular 3×3 matrices, $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{R}$, are such that

$$\mathbf{RA} = \mathbf{B}$$

The matrix $\mathbf{R}$ represents a rotation about the x-axis through an angle θ and

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\sin\theta & \cos\theta \\ 0 & \cos\theta & \sin\theta \end{pmatrix}$$

(a) Show that $\mathbf{A}$ is independent of the value of θ. **[3]**

(b) Give a full description of the single transformation represented by the matrix $\mathbf{A}$. **[1]**

Solution

(a) Since

$$\mathbf{RA} = \mathbf{B}$$

and $\mathbf{R}$ is non-singular, we can multiply on the left by $\mathbf{R}^{-1}$:

$$\mathbf{A} = \mathbf{R}^{-1}\mathbf{B}$$

For a rotation about the x-axis through angle θ,

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

so

$$\mathbf{R}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}$$

Now calculate $\mathbf{A}$:

$$\begin{aligned} \mathbf{A} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\sin\theta & \cos\theta \\ 0 & \cos\theta & \sin\theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\cos\theta\sin\theta + \sin\theta\cos\theta & \cos^2\theta + \sin^2\theta \\ 0 & \sin^2\theta + \cos^2\theta & -\sin\theta\cos\theta + \cos\theta\sin\theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \end{aligned}$$

This matrix contains no θ, so $\mathbf{A}$ is independent of the value of θ.

Hence

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

(b) The matrix $\mathbf{A}$ maps

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x \\ z \\ y \end{pmatrix}$$

so it leaves x unchanged and swaps y and z.

Therefore, $\mathbf{A}$ represents a reflection in the plane $y = z$.

11. **(a)** Specify fully the transformation $\mathbf{T}$ of the plane associated with the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

and $0 < \theta < 2\pi$. **[2]**

(b) (i) Find $\det \mathbf{M}$. **[1]**

(ii) Deduce **two properties** of the transformation $\mathbf{T}$ from the value of $\det \mathbf{M}$. **[2]**

(c) Prove that

$$\mathbf{M}^n = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix}$$

where n is a positive integer. **[4]**

(d) Hence specify fully a **single** transformation which is equivalent to n applications of the transformation $\mathbf{T}$. **[1]**

Solution

(a) For a point with position vector $\begin{pmatrix} x \\ y \end{pmatrix}$, the transformation gives

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

This is the standard matrix for a rotation in the plane.

So $\mathbf{T}$ is a rotation through angle θ anticlockwise about the origin.

(b) (i)

$$\det \mathbf{M} = \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = \cos\theta\cos\theta - (-\sin\theta)(\sin\theta)$$

$$= \cos^2\theta + \sin^2\theta = 1$$

(ii) Since $|\det \mathbf{M}| = 1$, the area scale factor is 1.

So areas are preserved.

Also, $\det \mathbf{M} > 0$, so orientation is preserved.

Therefore there is no reflection.

The two properties are: area is preserved, and orientation is preserved.

(c) We prove the result by induction on n.

First, when $n = 1$,

$$\mathbf{M}^1 = \mathbf{M} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos(1\theta) & -\sin(1\theta) \\ \sin(1\theta) & \cos(1\theta) \end{pmatrix}$$

So the statement is true for $n = 1$.

Now assume that for some positive integer k,

$$\mathbf{M}^k = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix}$$

Then

$$\mathbf{M}^{k+1} = \mathbf{M}^k\mathbf{M}$$

Substituting the induction hypothesis,

$$\mathbf{M}^{k+1} = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Multiplying the matrices,

$$\mathbf{M}^{k+1} = \begin{pmatrix} \cos(k\theta)\cos\theta - \sin(k\theta)\sin\theta & -\cos(k\theta)\sin\theta - \sin(k\theta)\cos\theta \\ \sin(k\theta)\cos\theta + \cos(k\theta)\sin\theta & -\sin(k\theta)\sin\theta + \cos(k\theta)\cos\theta \end{pmatrix}$$

Using the angle addition formulae,

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

we get

$$\mathbf{M}^{k+1} = \begin{pmatrix} \cos((k+1)\theta) & -\sin((k+1)\theta) \\ \sin((k+1)\theta) & \cos((k+1)\theta) \end{pmatrix}$$

So the statement is true for $n = k+1$.

If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction

$$\mathbf{M}^n = \begin{pmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{pmatrix}$$

for all positive integers n.

(d) From part (c), n applications of $\mathbf{T}$ have matrix

$$\mathbf{M}^n = \begin{pmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{pmatrix}$$

This is the matrix of a rotation through angle $n\theta$ anticlockwise about the origin.

So the single equivalent transformation is a rotation about the origin through angle $n\theta$ anticlockwise.